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A Cost-Effective NILM Solution With Three-Point Labelling and Non-Causal Convolution Technique

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ABSTRACT

Although deep learning is increasingly promising in the field of Non-Intrusive Load Monitoring (NILM) these days, the high costs of data recording and labelling represent a significant challenge for the training of supervised models. To address this, a cost-effective sequence-to-points NILM solution is proposed, integrating three-point labelling with non-causal convolution techniques. The approach introduces a semi-automatic labelling framework for obtaining NILM three-point data, which provides a low-cost data collection and labelling solution for large-scale applications. Then, a novel loss function combining coordinate loss and confidence loss is developed to address the positional misalignment and negative sample confusion in sequence-to-points scenario in NILM. Furthermore, an advanced neural network architecture based on multi-scale non-causal temporal convolution techniques is designed to capture unique features and operational modes of different appliances. Experimental results on the UK-DALE dataset show that the proposed mixed loss function has an advantage over plain Mean Absolute Error (MAE) on the sequence-to-points occasion, and the novel network outperforms on all of the appliances, demonstrating its potential for practical NILM applications.

1 | Introduction

With the gradual increase in global energy demands alongside environmental and economic pressures, the need for efficient energy management has become a common concern. Studies have shown that energy monitoring and direct feedback at the appliance level are highly beneficial [1]. In this context, technologies capable of analysing energy consumption patterns without additional infrastructural investments provide a pathway to manage and reduce energy expenditures [2, 3].

By installing load monitoring equipment at the household electric metre, rather than intrusively monitoring each appliance within a household, NILM employs power disaggregation algorithms to decompose the total power to extract the signals of the separate appliances. Efficient and accurate NILM can timely inform residents about detailed appliance-level power consumption, improving building operational efficiency at a relatively low installation cost. Thus, the effectiveness of NILM largely falls on the performance of the power disaggregation algorithms. Hart [4] first proposed a primitive event-based algorithm, which laid the groundwork for subsequent advancements. From then on, particularly with the advent of machine learning, many more algorithms have been developed.

These data-driven algorithms can be broadly divided into traditional machine learning and deep learning. In the former category, Factorial Hidden Markov Model (FHMM) [5, 6] and

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its multiple variants have been proven to be possible solutions since they take into consideration the probabilistic transitions between working states. Backed by the end-to-end convenience, Deep Neural Networks (DNN) have gradually occupied a dominant role in NILM, and have already made great achievements in related fields. Since Kelly et al. [7] introduced three pioneering DNN approaches, including Denoising Autoencoders (DAE), Bidirectional LSTM (BiLSTM), and a regression network, demonstrating their superiority over traditional FHMM and Combinatorial Optimisation (CO), a variety of deep learning techniques have emerged. DNN models also show promising prospects for real-time disaggregation, as they can be embedded into measurement devices as structured parameters after training [8].

Recently, DNN methods in NILM have been further categorised based on their output format: sequence-to-sequence (S2S), sequence-to-subsequence, and sequence-to-point (S2P), which define whether the output is a sequence of the same length as the input, a shorter sequence, or a single point, respectively [9]. Among S2S networks, variants of CNN dominate the majority, as shown by Herath et al. [10] and Garcia et al. [11]. Zhou et al. [8] enhanced these models with non-causal temporal convolutions and a multi-scale mechanism, proving that dilated convolution can improve performance with fewer parameters. There are many other original forms in terms of S2S, such as gated recurrent unit (GRU) and attention mechanisms, as explored by Murray et al. [12] and Piccialli et al. [13], have attempted to merge appliance state and power predictions. However, the entire sequence of inputs for each time step makes these S2S models computationally expensive for real-time disaggregation tasks.

To address the limitations of S2S, S2P was first introduced by [14], which combines predictions from different sliding windows, and achieves better identification results without the need for full sequence prediction. This approach has been further refined by combining temporal convolutions [15], binary regression techniques [16], and power estimation at adjacent points instead of midpoint within the window [17]. Although S2P occupies relatively low computational complexity, it tends to exhibit limited generalisation when dealing with uncommon devices or anomalous behaviours, particularly due to the requirement for large, high-quality, and diverse datasets.

Even though there are several public datasets available, the continuous demand for data with more device types and operational modes should not be ignored if further in-depth applications are needed. Meanwhile, the data labelling process is resource-intensive, both in terms of time and cost, involving extra equipment and considerable manual effort. In response to these limitations, researchers have explored different methods to improve the identification performance from the perspective of data. One approach is to generate synthetic data to expand the training set. Chen et al. [18] and Rafiq et al. [19] developed automatic synthesis algorithms based on statistical appliance operation patterns, but these methods lack the ability to model the behaviours of nonstandard or atypical power users. Klemenjak et al. [20] proposed a manual synthesis tool to generate more diverse virtual samples, yet the gap between synthetic data and real-world consumption distributions remains a challenge. Transfer learning aims to improve model generalisation when labelled data is scarce. For example, Michele et al. [21] demonstrated the proposed network shows promise for the transfer learning from a synthetic dataset to a real dataset, while Lin et al. [22] employed domain adaptation loss to minimise distributional differences between source and target domains, thereby improving performance with limited test data. However, it often requires extensive domain adaptation due to variations in appliance behaviours, which can be computationally expensive and challenging to implement in diverse environments. Besides, in terms of model scalability, Yin et al. [23] proposed an unsupervised domain adaptation framework to mitigate the generalisation degradation in heterogeneous energy consumption scenarios where data are not independently and identically distributed. Wang et al. [24] developed a transformer-based framework with controllable complexity and high transferability, demonstrating scalability across appliances and users. Virtsionis et al. [25] adopted a multi-target regression structure to build a unified model applicable to multiple appliances. These studies collectively shift NILM from single-appliance or user-specific models towards scalable frameworks that can adapt to multiple appliances and scenarios. Although such directions are valuable for practical deployment, they fall beyond the scope of this paper and will be discussed here.

Therefore, in terms of data acquisition and labelling, we also need to do some research to support large-scale practical application of NILM. In response to the existing challenges, this work makes the following contributions: 1. A simplified sequence-to-points learning occasion is proposed in this paper. The predicted appliance power sequence is replaced by three points, which are the turning-on time, the turning-off time point and the average power of the appliance. The occasion not only reduces the cost of labelled data collection in the early stage, but also optimises the identification process of model in scenarios where users only care about the energy consumption of the appliance's power cycle. In addition, we propose a low-cost semi-automatic labelling framework for obtaining NILM three-point data. 2. A novel loss function combining coordinate loss and confidence loss is developed, specifically designed to tackle the challenges in three-point regression scenarios. This innovation addresses the issues of misalignment between predicted values and ground truth, and confusion from negative samples, ensuring more accurate NILM performance. 3. An Advanced neural network architecture, Multi-Scale Non-Causal Temporal Convolutional Network (MSNCTCN), is designed to enhance NILM accuracy by employing non-causal temporal convolution and multi-scale feature extraction, enabling the model to capture distinctive operational features of appliances through an extended receptive field.

The remainder of this paper is organised as follows. Section 2 introduces the three-point labelling framework, the process of sample annotation generation, the loss function utilised, and the multi-scale non-causal temporal convolutional structures employed. Section 3 describes the experimental design, including data preparation, model implementation, and performance evaluation. Finally, Sections 4 and 5 conclude the paper and discuss future research directions.

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2 | Methodology

Figure 1 outlines the framework for the proposed NILM solution with three-point labelling and non-causal convolution techniques. The process starts with data acquisition and is followed by feature extraction and three-point label generation, culminating in training the MSNCTCN Network for improved appliance identification. A comprehensive explanation of components within this framework follows in the subsequent sections.

2.1 | Three-Point Labelling

Traditional sequence-to-sequence NILM focuses on disaggregating the appliance-level power sequences from the aggregated power data, where both inputs and outputs are sequences. In many scenarios, an abstract presentation of appliance load is adequate for meeting the needs of electricity users. Therefore, the three-point labelling method is proposed to achieve lightweight load identification. By compressing a complete power consumption sequence into three points: [turning-on time, turning-off time, average power], it not only significantly reduces recording redundancy during the off period of appliances, but also simplifies the data labelling process in this regression framework.

Within the three-point labelling framework, residents can act as potential contributors of sample labels. Considering that data quality is not strictly required, the residents can use the terminal application to record the turning-on and turning-off time of the appliance and manually report its rated power. According to this way, the actual equipment startup and shutdown detected by the event-based algorithms [26] in the main metre can be matched to the data annotations.

It should be noted that in the proposed three-point labelling method, both complete sub-metre data collected by nonintrusive devices and labels manually provided by residents are used to generate output labels during the model training phase. In actual deployment, the NILM method only relies on aggregated load collected by smart metres as input. Moreover, collecting labels by residents within this framework serves as a supplementary approach rather than a necessity for model training, the proposed method also supports public datasets or nonintrusive measurement data. In comparison, collecting incentivised data from residents through mobile applications (APPs) provided by state grid company is sometimes more cost-effective than installing additional hardware.

Assuming that the [turning-on time, turning-off time, average power] annotation information of all appliances has been obtained, a comprehensive framework can be designed for both training and testing prediction. In network training, taking the neural network designed for kettle detection as an example, the aggregated power sequence containing the complete operating cycle of the kettle and corresponding labels is added to the training set as positive sample, and the aggregated power sequence containing other appliances is set as negative sample, marked as [0,0,0]. In the actual prediction stage, suspicious on-off events of the appliances should be first located by the event detection algorithm. Then, the aggregated power sequence will be divided according to the usual operation cycle and sent to the neural networks to determine the appliance type and running time.

Although in ref. [9], such an occasion is categorised as sequence-to-subsequence, in the rest of the article, we still label it as S2P since the output is merely discrete points. In three-point regression method, it should be noted that the turning-on time always precedes turning-off time. In order to avoid confusion within the internal fitting architecture of neural networks, we modify these two values into the midpoint and the duration of an activation as shown in Figure 2. For the negative samples, all three points still have the same value of zero.

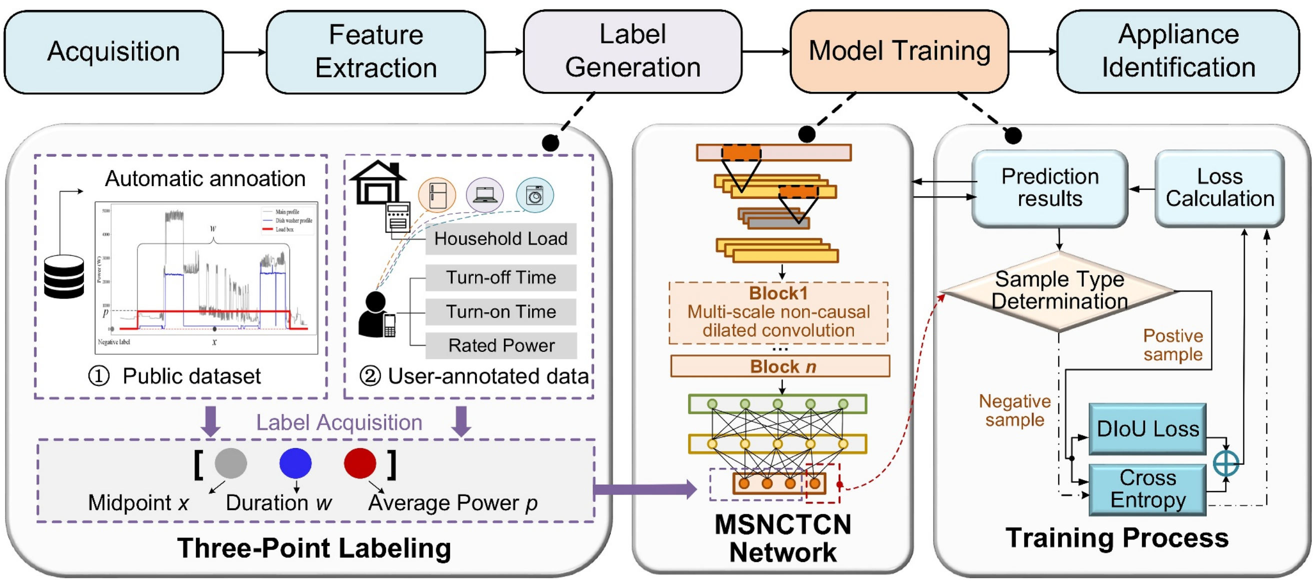

FIGURE 1 | Framework of proposed NILM Solution.

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2.2 | Sample Annotation Generation

This section analyses the feasibility of semi-automatic labelling using electrical appliances with both simple and complex working conditions as examples, and offers a comprehensive summary of the labelling method.

Take the dishwasher as a representative example. For electrical appliances with complex operating conditions, the dishwasher featuring multi-power gear is utilised as shown in Figure 3. During its operation cycle, there are both heating and washing processes with significant differences in power consumption between them. The event pairs 3, 4, 7 contain the heating process, while the other pairs contain the washing process. Although the label marked by the user only includes the user perception level information and does not include the intermediate process of the appliance operation, the heating operation interval can be located according to the equipment nameplate information, including auxiliary heating power reported by the user. For the washing process with small power, the concept of duty cycle can be used to describe it. Therefore, the final average operating power can be written as follows:

$$P_{avg} = \frac{\sum_{i=1}^{n} P_i \cdot T_i \cdot p_i}{T_{Label}} \tag{1}$$

P_i is the rated power of mode i, T_i is the operating time of mode i, and p_i is the operating duty cycle of mode i, T_{Label} is the application operating time marked by the user. For the dishwasher, the auxiliary heating is mode 1, and the corresponding duty cycle is 1. The flushing is mode 2, and the corresponding duty cycle is a certain value between [0,1](it is set to 0.5 in this experiment).

As shown in Table 1, we calculate the error between the actual average power of each application from sub-metre and the average power calculated by the above method from aggregate metre.

The results in Table 1 indicate that the proposed method of user collaborative marking has practical application value when data quality is not strictly required.

2.3 | IoU-Based Loss Function and Confidence Score

Usually, regression network uses mean squared error (MSE) as the loss function, but it applies more competently to a situation where the regressed values do not have geometrical or positional relations with each other. Although in our case,

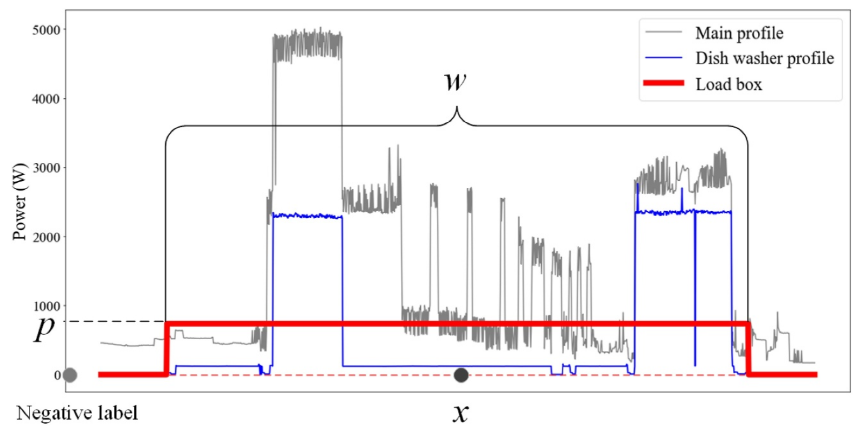

FIGURE 2 | A load box representing an activation of the dishwasher, characterised by midpoint x, duration w, and average power p.

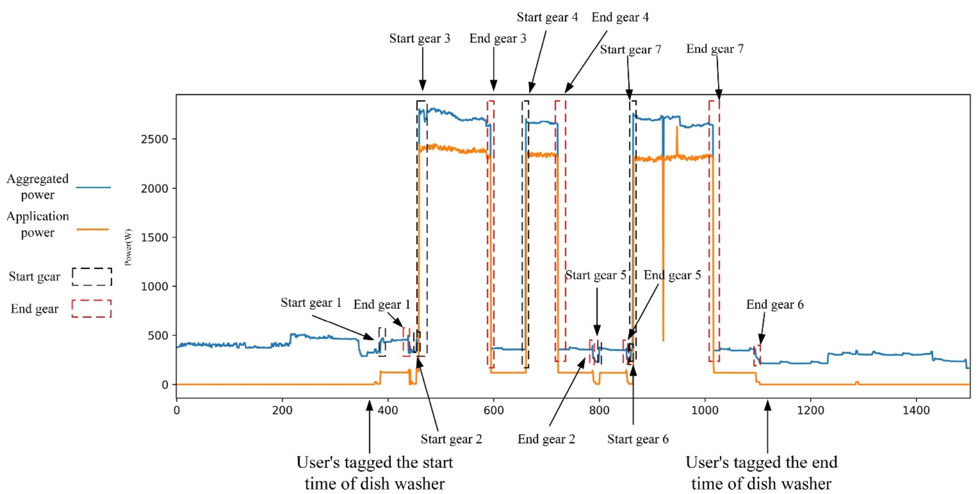

FIGURE 3 | Operation cycle of dish washer and corresponding aggregate power.

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TABLE 1 | The mean error between true average power and calculated average power.

Appliance	Mean error (W)	Standard deviation (W)	Deviation percentage
Kettle	15.344	2.213	0.557%
Microwave	31.213	4.213	1.225%
Dishwasher	35.514	5.937	1.663%
Washing machine	20.135	3.873	0.926%

they can form a simple rectangle resembling the bounding box in object detection. As Figure 1 shows, the bottom of the load box overlaps the time axis while the height is on power axis.

Because MSE cannot precisely indicate the correct relative positions between the prediction at this point and ground truth, we adopt intersection of union (IoU)-based loss functions to solve this problem, whose original form can be expressed as follows:

$$IoU = \frac{\mathrm{Pr} \cap Gt}{\mathrm{Pr} \cup Gt} \tag{2}$$

where Pr denotes the prediction and Gt denotes the ground truth. It is obvious that when two load boxes do not overlap, the IoU is a constant and cannot yield any gradient. In order to construct a loss function with stable gradients, GIoU (Generalised IoU) and DIoU (Distance IoU) are proposed to signify the exact distance between the two boxes. Among these two loss functions, DIoU further directly minimises the distance between two boxes as opposed to using areas, which is proven to be more precise and leads to faster convergence. It can be formulated as follows:

$$DIoU = IoU - \frac{d^2(O_{\mathrm{Pr}}, O_{Gt})}{c^2} \tag{3}$$

where $d(\cdot)$ is the Euclidean distance, and O denotes the midpoint of a load box, c denotes the diagonal of the convex C.

With the three-point labelling, the algorithm of calculating DIoU is shown in Algorithm 1.

ALGORITHM 1 | DIoU loss calculation.

Input: predicted midpoint and duration of an activation $\hat{x}, \hat{w}$, average power $\hat{p}$, and ground truth x, w, p
Output: L_{DIoU}
Step 1: Calculating IoU
$x_0 \leftarrow x - \frac{1}{2}w, \hat{x}_0 \leftarrow \hat{x} - \frac{1}{2}\hat{w}$
$x_1 \leftarrow x + \frac{1}{2}w, \hat{x}_1 \leftarrow \hat{x} + \frac{1}{2}\hat{w}$
$w_{\mathrm{inter}} \leftarrow \min(x_1, \hat{x}_1) - \max(x_0, \hat{x}_0)$
$w_{\mathrm{inter}} \leftarrow \max(0, w_{\mathrm{inter}})$
$p_{\mathrm{inter}} \leftarrow \min(p, \hat{p})$
$I \leftarrow w_{\mathrm{inter}} \cdot p_{\mathrm{inter}}, U \leftarrow wp + \hat{w}\hat{p} - I$
$IoU \leftarrow \frac{I}{U}$
Step 2: Calculating DIoU Loss
$d^2(O_{\mathrm{Pr}}, O_{\mathrm{Gt}}) \leftarrow (x - \hat{x})^2 + \frac{1}{4}(p - \hat{p})^2$
$c^2 \leftarrow (w + \hat{w} - w_{\mathrm{inter}})^2 + \max(p, \hat{p})^2$
$DIoU \leftarrow IoU - \frac{d^2(O_{\mathrm{Pr}}, O_{\mathrm{Gt}})}{c^2}$
$L_{\mathrm{DIoU}} \leftarrow 1 - DIoU$
return L_{DIoU}

L_{DIoU} has the range of $[0, 2]$, when achieving 0 means both boxes completely overlapping, achieving 2 means the two boxes are infinitely apart.

The problem with using IoU-based function as loss is that when encountering negative samples, the representation of the distance between the two load boxes will be confusing. With the negative labels all zeros, as Figure 2 shows, the L_{DIoU} can be calculated as follows:

$$L_{DIoU} = 1 - DIoU = 1 + \frac{\hat{x}2}{\left(\hat{x} + \frac{1}{2}\hat{w}\right)^2 + \hat{p}2} = 1 + \frac{1}{\left(1 + \frac{\hat{w}}{2\hat{x}}\right)^2 + \left(\frac{\hat{p}}{\hat{x}}\right)^2} \tag{4}$$

When the loss function is being minimised, it means minimising $\hat{x}$ while maximising $\hat{w}$ and $\hat{p}$, as for the L_{DIoU}, we cannot prompt the prediction to get close to ground truth. Thus, it is necessary to introduce a place for confidence score, by which the network can present how certain a belief is that one is a positive sample. The confidence score is in the range of $[0, 1]$, as in probability, can be minimised using cross-entropy function.

Taking all factors into consideration, the complete loss function for the proposed method is composed of two parts: coordinate loss and confidence loss. For a negative sample, only confidence loss would be calculated while for a positive one, both parts would be summed up. It can be written as follows:

$$\begin{aligned}\mathrm{Loss} = &\sum_{n=1}^{N} \mathbb{Z}_n^{\mathrm{pos}}(1 - DIoU) \\ &+ \sum_{n=1}^{N} \left[\mathbb{Z}_n^{\mathrm{pos}} \log(\hat{P}_n) + (1 - \mathbb{Z}_n^{\mathrm{pos}})\log(1 - \hat{P}_n)\right]\end{aligned} \tag{5}$$

Where N denotes the number of categories of appliances in one sample, $\mathbb{Z}_n^{pos}$ takes its value in a two-elements set $\{0, 1\}$, and equals to one only when the sample is positive for the nth appliance, and $\hat{P}_n$ denotes the confidence score for the nth appliance. In this paper, we train one network for each type of appliance, so $n = 1$.

2.4 | MSNCTCN

Here, we detail the structure of the proposed MSNCTCN and introduce other DNN structures adopted in the experiments in the next section.

Inspired by TCN [27] and multi-scale mechanism from [8, 28], MSNCTCN takes residual block as basic unit. The structure of a

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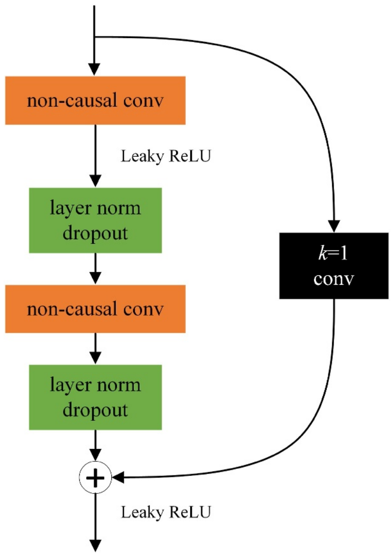

FIGURE 4 | Details of a residual block.

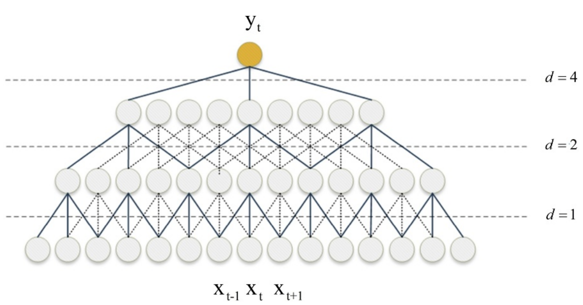

FIGURE 5 | Non-causal dilated convolution.

residual block is shown in Figure 4, consisting of two non-causal dilated convolutional layers with the same dilation rate and number of output channels. A shortcut connection is used to add up the input of a block to the output in order to directly pass features in shallower layers to the deeper ones, preventing the network from degradation when the number of layers increases. A $k = 1$ convolution is optional in the shortcut connection only to modify the number of channels of both input and output to be equal.

The details of non-causal dilated convolution are shown in Figure 5, d denotes the dilation rate. By filling $d - 1$ zeroes in between every two parameters of common kernels and exponentially increasing dilation rate with the network getting deeper, the convolution gains larger receptive fields to get a better learning capacity and avoids explosion of parameters. By taking both past and future information into account, the convolution obtains non-causality, which enriches the context information network needed when performing power disaggregation.

Given that different appliances and different working modes may have a wide range of features, multi-scale mechanism makes the network automatically pick the optimal receptive fields for specific features. Inspired by GoogLeNet [29], we break a big chunk of multi-scale structure into several multi-scale blocks, each block has three branches of different receptive fields and each branch has different numbers of residual blocks.

As shown in Figure 6, each branch first deploys a $k = 1$ convolution to compress the information and increase the nonlinearity in the feature maps. Two branches are built by connecting residual blocks, and the third uses $k = 1$

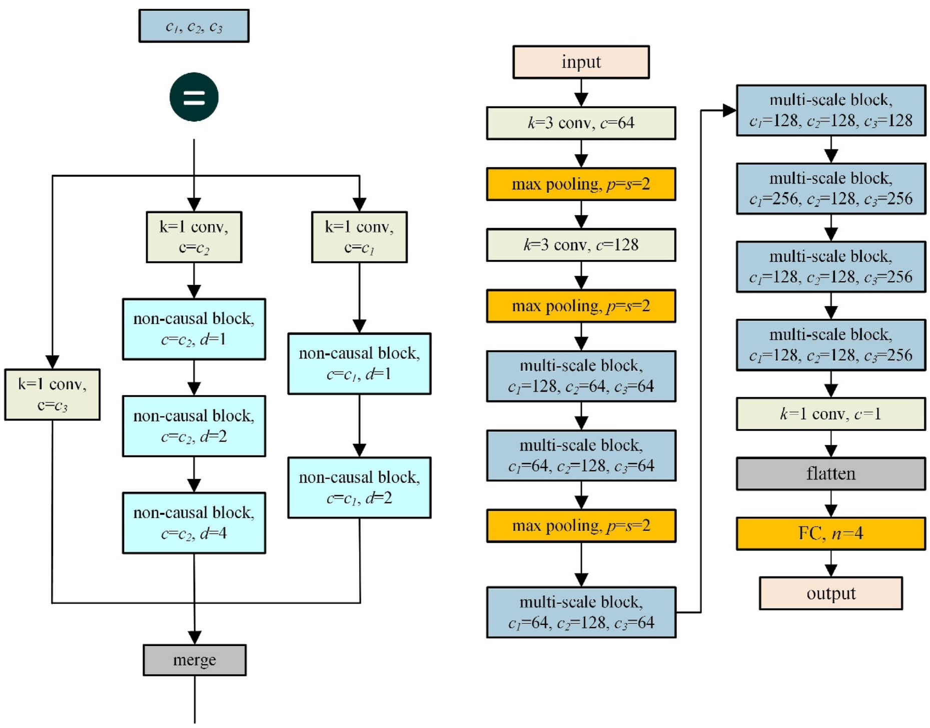

FIGURE 6 | Structure of MSNCTCN.

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convolution again to pass on the features from previous layers with another route.

The inputs firstly go through two initial convolutions and their following max pooling layers (*p* denotes pool size, *s* denotes stride), then go through two consecutive multi-scale blocks, a max pooling layer, and another five consecutive multi-scale blocks. After the number of channels is reduced to 1, a fully-connected layer is used to present the output with a sigmoid function.

3 | Experiments

3.1 | Dataset: UK-DALE

The public NILM dataset UK-DALE [30] is chosen as the dataset for our experiment. In this case, five houses were monitored for power sequence data, and most of them were recorded every 6 seconds. Additionally, considering the common usage and power consumption ratio of household appliances, we have chosen the kettle, microwave oven, dishwasher, and washing machine as target appliances for experiments. Even though the fridge is also energy-intensive, its steady operating pattern makes its load substantially easier to monitor compared to other appliances. Meanwhile, the dataset is compatible with the NILMTK project [31], and one of the functions in NILMTK (*get_activation*()) can extract activations from sub-metre data in list form, and the UK-DALE dataset provides the parameters required for this function.

In order to validate the effectiveness of the proposed annotation method in this paper, a simulation of annotation was conducted on the dataset, and the UK-DALE dataset provides the parameters required for this function. The *get_activation* function was utilised to obtain device operation information, such as the turning-on and turning-off times from submetres. This information was then used to simulate the annotated information by users, enabling the computation of average device power within the corresponding intervals in the aggregate data. Besides, in real-world scenarios, user annotations may be inconsistent because the recorded activation may be shifted relative to the actual device behaviour. In this study, manual annotations, automatic annotations, and simulated manual annotations are integrated within a unified training set generation framework. In Algorithm 2, a tolerance mechanism refines user-annotated segments via a sliding-window search guided by a pretrained binary discriminator. The search adjusts timestamps within a temporal neighbourhood to better capture the appliance's true power behaviour, discarding samples when no reliable interval is found. Given the minor discrepancy between simulated user annotations and actual labels and the inevitable inaccuracy of user-provided operation data in real scenarios, the simulation results and actual labels were used as two separate label sets for comparison in this experiment.

Meanwhile, considering the practical application of the proposed model in this paper, the sample generation needs to follow a certain rule, which is consistent with the data slicing method in actual prediction. As mentioned in Section 2, in the practical identification, the turning-on event of the suspicious appliance will be firstly detected according to the event-based algorithm, and then the execution of the NILM model will be triggered. The input sequence of the neural network will generally be intercepted from a certain time before the appliance starts. However, considering the impact of load fluctuation in the metre, there may be deviations in the starting time of the appliances detected in the real scene. Therefore, the first data of the sample starts 1–2 min randomly before the event. When selecting the sample window length, the model inputs for training different devices are inconsistent. Positive samples of a certain appliance type are extracted according to the parameters in Table 2. An event that lasts longer than Min. On duration can be considered to be a valid event while two valid events, of which the blank interval is longer than Min. off duration can be considered to be two valid events; otherwise, they are merged into one valid event. Only when a valid event meets all the parameter requirements can it be selected as a positive sample. When selecting negative samples of a certain type of appliance, turning-on events of nontarget appliances are located and data cutting is performed according to the window length of this type.

ALGORITHM 2 | Target application's training set generation.

Input: Timestamp set Γ of activation events (includes $1:1$ $\Gamma_{\text{TargetApp}}$ and $\Gamma_{\text{NotTargetApp}}$), aggregate sequence M, max aggregate power $P_{\max}$, sample window length s, context offset t_{offset}
if *user annotation* **then**
  Rated power P_{rated}, temporal offset tolerance δ
  Pretrained discriminator $f(\cdot)$, reliable threshold σ
else if *automatic or simulated user annotation* **then**
  Submetred power sequence N
end if
Output: Sequence set Φ_{seq}, label set Φ_{label}
Step 1: Refined Positive Sample Generation
for all *act* in first 90% of $\Gamma_{\text{TargetApp}}$ **such that** $len(act) > s$ **do**
  $t_0 \leftarrow act.start$, $t_1 \leftarrow act.end$
  if *user annotation* **then**
    $best\_score \leftarrow \sigma$, $refined\_found \leftarrow$ False
    for $t_{\text{start}} \in [t_0 - \delta,\ t_0 + \delta]$ **do**
      for $t_{\text{end}} \in [t_1 - \delta,\ t_1 + \delta]$ **do**
        if $t_{\text{end}} - t_{\text{start}} < s$ **then**
          continue
        end if
        $seq \leftarrow M[t_{\text{start}} : t_{\text{start}} + s]$
        $score \leftarrow f(seq)$
        if $score > best\_score$ **then**
          $best\_score \leftarrow score$
          $refined\_t_0 \leftarrow t_{\text{start}}$, $refined\_t_1 \leftarrow t_{\text{end}}$
          $refined\_found \leftarrow$ True
        end if
      end for
    end for
    if $refined\_found$ **then**
      $t_0 \leftarrow refined\_t_0$, $t_1 \leftarrow refined\_t_1$
    else
      continue ▷ Skip unreliable annotation
    end if
  end if

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Compute P_{avg} from P_{rated}
    $seq \leftarrow M[t_0 - t_{\text{offset}} : t_0 - t_{\text{offset}} + s]/P_{\max}$
    $label \leftarrow [t_0 - t_{\text{offset}},\ t_0 - t_{\text{offset}},\ P_{\text{avg}}/P_{\max}]$
    $confidence \leftarrow 1.0$
    Add seq to Φ_{seq}; Add $(label, confidence)$ to Φ_{label}
end for
Step 2: Negative Sample Generation
for all act in first 90% of $\Gamma_{\text{NotTargetApp}}$ **such that** $len(act) > s$ **do**
    $t_0 \leftarrow act.start$
    $seq \leftarrow M[t_0 - t_{\text{offset}} : t_0 - t_{\text{offset}} + s]/P_{\max}$
    $label \leftarrow [0,\ 0,\ 0]$
    $confidence \leftarrow 0.0$
    Add seq to Φ_{seq}; Add $(label, confidence)$ to Φ_{label}
end for
return $\Phi_{\text{seq}}, \Phi_{\text{label}}$

Algorithm 2 also shows the process of generating positive and negative samples for different appliances, where the ratio of positive and negative samples is 1:1. It is worth noting that in this algorithm, Φ_{label_0} represents the label dataset obtained by the automated annotation method using the semi-automatic annotation approach from the aggregate data, while Φ_{label_1} represents the actual label dataset corresponding to the sub-metre data. Meanwhile, after the normalisation process, each sample and its mean value will be scaled to [0, 1]. Then, 90% of the extracted activations from each house are used to generate the training set, and the remaining 10% (except for the data of microwave and washing machine in house 4, which are discarded because they share one electricity metre when recorded) form the test set.

3.2 | Network Model and Training Process

We have chosen two DNNs to perform the comparison experiment: the most common S2P network used to regress the midpoint power [14] and the customised state-of-the-art S2S WaveNILM network [32]. The structures of the network are shown in Figure 7. The activation function of output layer for all models is sigmoid function.

When using the simulated annotation-generated data as the label dataset to conduct experiments, the model takes the extracted aggregate sequence data as input, with the corresponding simulated user-annotated data as the comparative labels. Conversely, when using the three-point labels generated from real sub-metre data sequences to conduct experiments, the model still takes the extracted aggregate sequence data as input, with the actual labels as the comparative labels. Through these two sets of experiments, the quality of the data produced by the annotation method and the performance of the proposed model in this paper can be evaluated.

Additionally, during the training process of the network models in both sets of experiments, we use a batch size of 32, and select the ADAM optimiser with an initial learning rate of 0.0001 for adaptive learning during the training process. Experiments are conducted on a computer equipped with an Intel Core i9-9900K CPU and an NVIDIA RTX TITAN 24 GB GPU.

3.3 | Results and Analysis

We conduct cross-experiment using both MSE loss function and L_{DIoU} loss functions on all four appliances with four networks.

There are two evaluation metrics we adopt for the experiments. The first one is Estimation Accuracy (Est.Acc), a metric to uniformly evaluate the regression accuracy of the disaggregation across different power levels of different appliances. It is calculated as follows:

$$\text{Est.Acc} = 1 - \frac{\sum_{t=1}^{T} |\hat{p}_t - p_t|}{2\sum_{t=1}^{T} p_t} \tag{6}$$

where p_t denotes the ground truth at time t, $\hat{p}_t$ denotes the disaggregated value at time t. While in order to avoid the noises during appliance's off period, a square-wave power sequence is firstly restored from the points label for S2P before calculating estimation accuracy. Other than that, we also adopt Mean Absolute Error (MAE) to evaluate the performance in a more specific manner:

$$\text{MAE} = \frac{\sum_{t=1}^{T} |\hat{p}_t - p_t|}{T} \tag{7}$$

In Tables 3 and 4, we present the identification results of the four household appliances using the three-point label data generated from both the simulated user-annotated data and the real sub-metre data sequences as experimental labels. Additionally, Table 5 shows the errors between the predicted values of the models in the two sets of experiments. Figure 8 displays some sample results from the best-performing model with the corresponding ground-truth labelled datasets in the sub-metre.

From what the results show, IoU-based loss function indeed has profound improvements on three-point labelling occasion. Both

TABLE 2 | Arguments passed to get_activation (), maximum power and window size for each appliance.

Parameter	Kettle	Microwave	Dishwasher	Washing machine
On threshold (W)	2000	200	10	20
Min.on duration (s)	12	12	1800	1800
Min.off duration (s)	0	30	1800	160
Max power (W)	3100	3000	2500	2500
Window size	64	64	1500	1024

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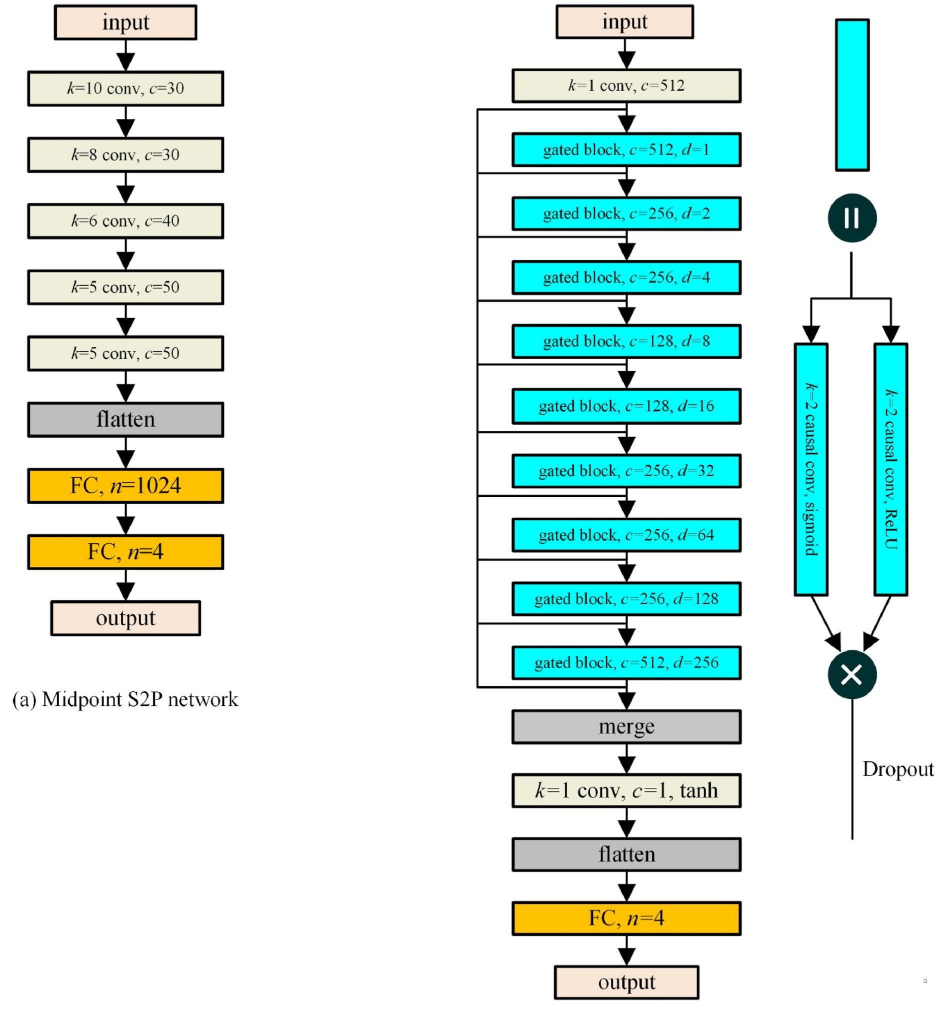

(a) Midpoint S2P network

(b) Customized WaveNILM network

FIGURE 7 | Structure of midpoint S2P network and customised WaveNILM network.

TABLE 3 | Results of NILM using automatically labelled generated data as labels.

		Kettle		**Microwave**		**Dish washer**		**Washing mach**	
Metric	**Model**	**DIoU**	**MSE**	**DIoU**	**MSE**	**DIoU**	**MSE**	**DIoU**	**MSE**
Est.Acc.	S2P	0.968	0.957	0.899	0.880	0.885	0.845	0.817	0.803
	WaveNILM	0.968	0.962	0.885	0.888	0.895	0.859	0.815	0.809
	MSNCTCN	**0.969**	0.963	**0.912**	0.896	**0.917**	0.893	**0.823**	0.812
MAE	S2P	34.918	48.232	81.256	96.431	71.329	96.245	87.188	94.062
	WaveNILM	35.899	43.062	92.461	90.453	65.431	87.209	88.345	91.093
	MSNCTCN	**34.966**	41.364	**71.083**	83.486	**51.901**	66.611	**84.516**	90.032

Note: The bold values represent the performance metrics of the proposed method in this paper.

MSNCTCN and WaveNILM achieve better estimation accuracy with L_{DIoU} than methods with common MSE on all four appliances. Some performance gaps, as with MSNCTCN on dish washer and WaveNILM on kettle, can be enormous. On the other hand, performance with L_{DIoU} maintains consistency across different appliances, proving its applicability under different situations. Figure 8 also shows that on right occasions, L_{DIoU} has an advantage in terms of quantifying the distance between predictions and labels, thus driving the network to find the optimal parameters.

Among these appliances, kettle has the best estimation accuracy results, since the profile of kettle is the simplest of them all, so the networks can easily learn the features of a square wave. On the contrary, dish washer and microwave have less satisfying results. For dish washer, the duration of

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TABLE 4 | Results of NILM using true labelled data as labels.

Metric	Model	Kettle		Microwave		Dish washer		Washing mach	
		DIoU	MSE	DIoU	MSE	DIoU	MSE	DIoU	MSE
Est.Acc.	S2P	0.981	0.976	0.906	0.891	0.961	0.930	0.797	0.755
	WaveNILM	0.979	0.975	0.880	0.853	0.955	0.929	0.820	0.662
	MSNCTCN	**0.982**	0.976	**0.916**	0.911	**0.963**	0.947	**0.855**	0.851
MAE	S2P	25.372	32.735	46.276	53.634	18.897	34.348	37.399	45.167
	WaveNILM	28.583	33.717	58.793	72.031	21.955	34.498	33.205	62.179
	MSNCTCN	**25.230**	33.095	**34.369**	43.797	**18.113**	25.750	**26.753**	27.472

Note: The bold values represent the performance metrics of the proposed method in this paper.

TABLE 5 | Comparison of prediction mean errors between two sets of experimental network models.

Appliance	S2P	WaveNILM	MSNCTCN
Kettle	11.115	13.211	10.143
Microwave	35.341	36.325	35.232
Dishwasher	45.127	44.242	44.202
Washing machine	48.892	47.347	43.935

an activation can be very long and the peak-valley difference is huge, and both of these factors contribute to the difficulties of nailing the exact coordinates of a load box. As for microwave, an uncertain amount of 'turning-off' time during an activation can lead to confusion between positive and negative samples.

When it comes to network performance with L_{DIoU} specifically, our MSNCTCN outperforms other networks on all appliances in estimation accuracy. The midpoint S2P network composed of consecutive traditional convolutional layers only outperforms WaveNILM on kettle by a small margin. It shows that simple structure may slightly improve performance of disaggregating appliances with simple activation features, but it still has disadvantages when processing more difficult tasks. Comparing MSNCTCN with WaveNILM, it proves that non-causality contributes more to regressing the coordinates of load boxes than causality by setting its receptive fields both on past and future inputs to expose the layer to a larger amount of information. Besides, MSNCTCN consistently outperforms WaveNILM on every appliance, which proves that the multi-scale structure has the advantage of automatically learning features on different scales for different appliances.

Finally, according to the result presented in Table 5, the comparison of prediction errors between the two sets of experiments reveals that when using simulated user-annotated data as label data, although some prediction errors are introduced, they remain within an acceptable range for the four electrical appliances. This demonstrates that within the framework of automated annotation, using user-assisted annotation methods without strict requirements for label data quality can still ensure the acquisition of accurate power consumption information even with reduced data complexity. In settings where the primary interest is in overall energy consumption rather than in fine-grained usage details, such as in many residential and commercial scenarios, this approach proves adequate for effective energy management.

3.4 | Sensitivity Analysis

To evaluate the potential impact of enhanced signal noise and nonstandard user behaviours on model performance in real-world applications, this section simulates complex input conditions to examine the robustness of the proposed method.

In practical scenarios, power signals are often affected by sampling accuracy, external electromagnetic interference, which introduces varying levels of noise into the collected data. To simulate such conditions, zero-mean Gaussian noise with different intensities was injected into the aggregate power signals from the UK-DALE training set. Let the aggregate power be denoted as $\mathbf{x} = \{x_t\}_{t=1}^{T}$ and the ground-truth power of the target appliance be $\mathbf{y} = \{y_t\}_{t=1}^{T}$. The noise perturbation is formulated as follows:

$$\mathbf{x}' = \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(0, \sigma^2 \mathbf{I}\right)$$

To ensure consistency in scale, the power data are normalised to the range [0,1], and the standard deviation σ is set to $\{0, 0.05, 0.1\}$. The perturbed input $\mathbf{x}'$ is then fed into the models, and the Mean Absolute Error (MAE) is used as the evaluation metric to compare the performance of models trained with different loss functions.

As shown in Table 6, as the noise intensity increases, MAE rises across all models, indicating degraded performance. However, the proposed MSNCTCN model consistently demonstrates stronger robustness across all noise levels, with a slower error growth rate compared to the baseline models. Furthermore, models trained with DIoU loss generally outperform those trained with traditional MSE loss, highlighting the advantage of the proposed method under noisy conditions.

In real-world usage, user behaviour often deviates from idealised operation modes, such as premature device shutdown or sudden operation mode changes. To simulate the impact of these nonstandard behaviours and reflect practical data conditions, we introduce a sequence truncation perturbation experiment, where incomplete operation segments are constructed to assess model performance.

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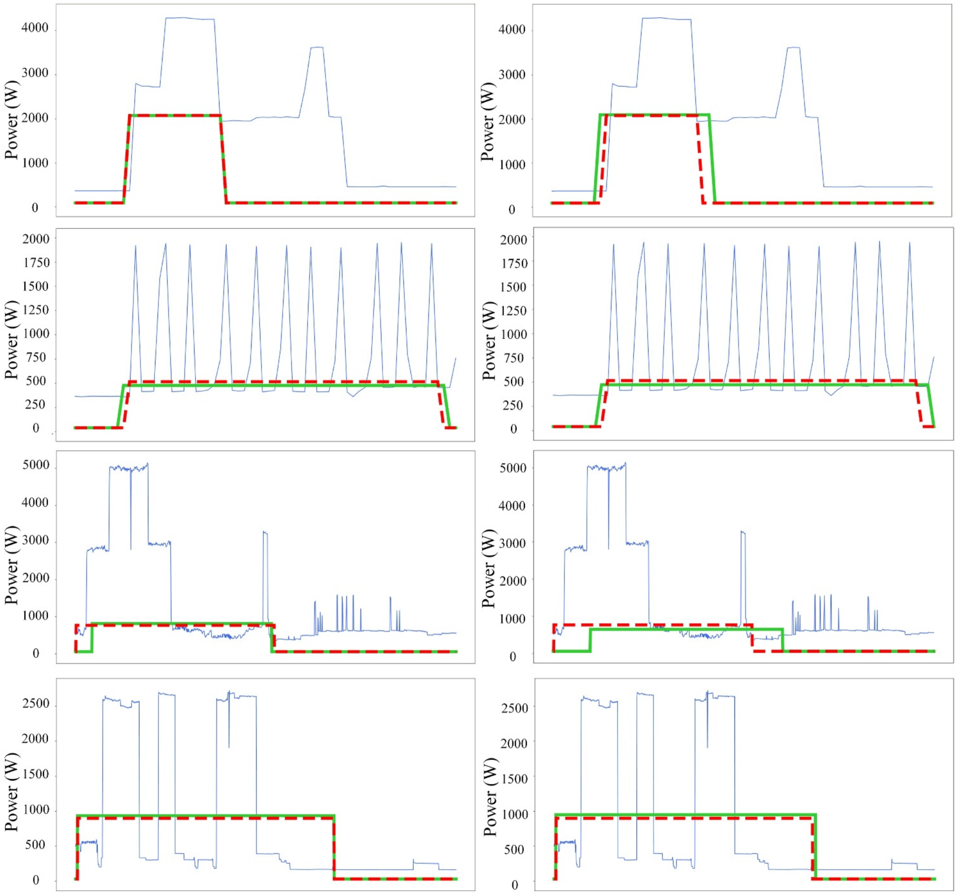

FIGURE 8 | Sample results on S2P occasion. From top to bottom: kettle, microwave, washing machine, dish washer. First three appliances are predicted by MSNCTCN, dish washer is predicted by WaveNILM. Left column uses L_{DIoU}, right column uses MSE. The red dash-line denotes the ground truth, the green solid line denotes the prediction.

TABLE 6 | MAE performance of different models under varying Gaussian noise levels.

		Kettle		**Microwave**		**Dish washer**		**Washing mach**	
σ	**Model**	**DIoU**	**MSE**	**DIoU**	**MSE**	**DIoU**	**MSE**	**DIoU**	**MSE**
0	S2P	34.918	48.232	81.256	96.431	71.329	96.245	87.188	94.062
	WaveNILM	35.899	43.062	92.461	90.453	65.431	87.209	88.345	91.093
	MSNCTCN	**34.966**	41.364	**71.083**	83.486	**51.901**	66.611	**84.516**	90.032
0.05	S2P	35.810	46.018	83.462	91.147	72.207	93.645	90.667	99.234
	WaveNILM	36.142	44.370	95.870	88.452	68.365	83.306	91.754	102.135
	MSNCTCN	35.092	42.105	73.458	86.707	55.873	71.492	89.107	96.923
0.1	S2P	37.810	49.612	89.382	98.648	78.462	101.663	95.907	113.951
	WaveNILM	38.489	47.069	98.707	95.340	72.000	95.244	96.180	120.177
	MSNCTCN	37.063	46.582	78.191	93.707	60.090	78.008	95.968	105.485

Note: The bold values represent the performance metrics of the proposed method in this paper.

We randomly select 10% of the samples from the test set and apply random truncation to the aggregate and target device power sequences during their operation intervals. For each selected sequence $\mathbf{x} = \{x_t\}_{t=1}^{T}$ and its corresponding target sequence $\mathbf{y} = \{y_t\}_{t=1}^{T}$, a random truncation point t_c is selected within the active interval, and truncated sequences are constructed as follows:

$$\mathbf{y}_t' = \begin{cases} y_t, & t \le t_c \\ 0, & t > t_c \end{cases}, \quad \mathbf{x}_t' = x_t - (y_t - y_t')$$

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TABLE 7 | MAE performance under normal and truncated sequence conditions.

Scenario	Model	Kettle		Microwave		Dish washer		Washing mach	
		DIoU	MSE	DIoU	MSE	DIoU	MSE	DIoU	MSE
Normal	S2P	34.918	48.232	81.256	96.431	71.329	96.245	87.188	94.062
	WaveNILM	35.899	43.062	92.461	90.453	65.431	87.209	88.345	91.093
	MSNCTCN	**34.966**	41.364	**71.083**	83.486	**51.901**	66.611	**84.516**	90.032
Trunc. 10%	S2P	35.269	46.121	85.913	92.887	72.522	92.343	94.268	98.201
	WaveNILM	36.306	44.215	94.556	91.607	68.605	85.147	90.549	97.801
	MSNCTCN	34.571	42.582	72.703	86.844	55.351	70.929	89.081	95.402

Note: The bold values represent the performance metrics of the proposed method in this paper.

The modified $\mathbf{x}'$ is used as model input, and $\mathbf{y}'$ as the reference label. The resulting MAE performance under truncated conditions is reported in Table 7.

Results show that short-duration high-power devices (e.g., kettle, microwave) are less affected by truncation, while long-duration, low-power devices (e.g., dish washer, washing machine) experience more noticeable performance degradation. This may be due to incomplete operation segments, making it harder for the model to recognise device states or treat low-power segments as invalid. Nevertheless, the MSNCTCN model consistently exhibits less performance drop, and DIoU-based models outperform their MSE-driven counterparts.

4 | Discussion

This study proposes a cost-effective sequence-to-points NILM solution based on a three-point labelling method, and designs the Multi-Scale Non-Causal Temporal Convolutional Network combined with a DIoU loss function. Experimental results on the UK-DALE dataset demonstrate that the proposed method outperforms representative models, achieving a favourable balance between efficiency and accuracy. Through a systematic evaluation, we observe that the model exhibits strong boundary localisation capabilities across different appliance types. In particular, for complex devices characterised by long operating cycles and low-power stages, the DIoU-driven model achieves superior prediction accuracy compared to conventional MSE-driven models.

To further evaluate the robustness of the method under noise interference and behavioural uncertainty in real-world environments, two representative experimental settings are introduced. First, Gaussian noise of varying intensities is injected into the original aggregate power data to simulate measurement errors and external interference. Results show that, although all models exhibit increased errors under noisy conditions, the DIoU loss preserves a significantly smaller MAE increase than MSE-based models, and the prorposed MSNCTCN model consistently demonstrates stronger robustness across all noise levels, demonstrating better stability. Second, we randomly truncate the operation cycles of some training samples to simulate user-interrupted operations and abnormal usage patterns. The results further indicate that the proposed method experiences a smaller performance drop under such nonstandard behaviours than other baseline methods, suggesting stronger generalisation capability.

Despite its promising performance, this study has several limitations. First, users' three-point label construction relies on submetre data and NILMTK tools. Although feasible under controlled conditions, how to reliably validate real-world user labels remains an important future work. Second, the current experiments are conducted on the UK-DALE public dataset. Although this dataset has certain representativeness, it cannot comprehensively capture regional differences in electrical infrastructure, appliance diversity, and user behavioural complexity. Therefore, the model's cross-domain transferability remains to be further validated. Additionally, in some overlapping-event scenarios, we observed boundary detection offsets, which may affect accuracy when multiple devices operate simultaneously.

Future work will focus on enhancing the adaptability and deployability of the model. On one hand, we plan to incorporate more types of data and real-world user-labelled data to validate the model's generalisation capability across regions and appliance categories. On the other hand, considering the need for edge deployment, we will investigate multi-task-based model integration and transfer-learning-based fine-tuning techniques to support and improve its adaptability in dynamic environments. We believe that the sequence-to-points paradigm based on three-point abstraction, combined with a robust and efficient model integration framework, offers a more practical pathway for nonintrusive load monitoring tasks.

5 | Conclusion

In many residential and commercial scenarios, an abstract representation of appliance loads is sufficient for meeting the needs of electricity users. In settings where the primary interest is in overall energy consumption rather than in fine-grained usage details, we propose a cost-effective sequence-to-points NILM solution.

To address the growing challenge of insufficient training data in deep learning, this paper proposes a low-cost three-point labelling method, replacing conventional appliance power sequences with three labels: turning-on time, turning-off time, and average power. A semi-automatic framework for acquiring three-point labels is also presented, enabling training data to be

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collected not only through submetred NILM devices but also through active participation of residential users, thereby offering a supplemental, installation-free labelling approach. Building on this method, we designed the MSNCTCN model based on non-causal convolution to fully exploit its extended temporal context. Furthermore, a novel DIoU loss function combining regression loss with confidence scoring is introduced to mitigate positional misalignment in three-point regression. Experimental results confirm the feasibility of the proposed framework and demonstrate superior performance over existing benchmarks.

For future work, we will focus on two directions to enhance scalability: (1) Collaborating with real users to collect three-point labelling data to validate method adaptability in real-world scenarios. (2) Focusing on potential network redundancy arising from training separate models for each appliance, we propose a lightweight architecture based on transfer learning and a multi-task joint modelling framework. The design mainly utilised shared feature extractors and task-specific branches to maximise parameter reuse and enable flexible task switching, thereby improving system scalability and deployment efficiency.

Author Contributions

Yanan Zhang: methodology, formal analysis, investigation, software, writing — review and editing. **Gan Zhou:** supervision, conceptualisation, writing — review and editing. **Yanjun Feng:** data curation, investigation, validation. **Zhan Liu:** methodology, software development, visualisation. **Li Huang:** data processing, results analysis. **Zhi Li:** software, technical support, writing — original draft. **Rui Bo:** supervision, writing — review and editing.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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